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TRA-SAE: A Low-Cost Curriculum and Reward-Shaping Study for Mixed-Domain Scientific Reasoning with Small Language Models

ANONYMOUS AUTHOR(S)[1]

[1][TODO: Affiliation, City, Country (e-mail: xxx@xxx.edu)]

Corresponding author: [TODO: Name] (e-mail: [TODO: email]).

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ABSTRACT Scientific question answering requires both numerical physical reasoning and symbolic logical inference, yet resource-constrained settings often preclude large proprietary models or extensive reinforcement learning. This paper presents TRA-SAE (Training with Reward-Augmented Self-consistent Adaptive Evaluation), a low-cost curriculum training pipeline for the EXACT 2026 mixed-domain scientific QA task using a 4B open model. TRA-SAE combines general supervised fine-tuning (SFT), a logic-focused curriculum adaptation stage, and GRPO with lightweight format, correctness, and unit-consistency reward shaping—total training cost \$18, full pipeline under \$64 on a single A100-80GB. On a 217-sample validation split, the best single configuration improves Qwen3.5-4B from 35.48% to 53.92% overall accuracy (67.38% physics, 28.95% logic), with the dominant gain of +17.05 pp attributable to curriculum SFT alone. Multi-seed evaluation confirms $53.15 \pm 1.15\%$ overall accuracy for the GRPO configuration across three seeds. Reward ablations show that correctness-only training converges faster under short ($\leq$150-step) budgets and achieves higher single-number accuracy than the full reward mixture, suggesting auxiliary unit-consistency and format signals require longer training to provide net benefit. Negative results reveal that self-consistency voting and Dual-LoRA expert routing degrade overall performance at 4B scale, attributable to systematic (non-random) reasoning errors and noisy domain-boundary routing respectively. A structured error taxonomy identifies logical-fallacy and quantifier errors (55.3% of wrong predictions) as the primary bottleneck, indicating that formal logic at this scale requires stronger symbolic supervision beyond standard SFT and GRPO. These findings offer practical guidance for scientific QA under strict compute constraints. All code, checkpoints, and evaluation scripts are released.

INDEX TERMS Large language models, scientific reasoning, reinforcement learning from human feedback, group relative policy optimization, curriculum fine-tuning, physics question answering, logical reasoning, parameter-efficient fine-tuning, LoRA

I. INTRODUCTION

Large language models (LLMs) have demonstrated remarkable capabilities on mathematical and logical reasoning benchmarks [1]–[3]. However, deploying capable models in resource-constrained settings—where larger proprietary models such as GPT-4 are unavailable—requires careful co-design of architecture, data curriculum, and reward specification.

Scientific reasoning occupies a particularly challenging niche: physics problems require formula selection, symbolic manipulation, and unit-consistent numerical computation, while logic problems demand quantifier handling, propositional inference, and recognition of common fallacies. These two modalities benefit from different inductive biases, yet a single model is expected to handle both in practice.

The EXACT 2026 competition provides a controlled testbed with 1,945 training examples (1,213 physics, 732 logic) and a hidden evaluation set, with a public baseline at 38.71%. We treat this competition as a proxy for real-world mixed-domain scientific Q&A and investigate how far a Qwen3.5-4B model can be pushed via publicly available fine-tuning techniques.

Our main contributions are:

- Low-cost curriculum pipeline: a three-stage training procedure (general SFT → logic curriculum SFT → GRPO reward shaping) for mixed-domain scientific QA using a 4B open model, with full end-to-end cost under $64 on a single A100-80GB.
- SFT-dominant empirical finding: curriculum SFT contributes the largest single gain (+17.05 pp over zero-shot), while GRPO adds a smaller top-end improvement (+1.39 pp); the GRPO gain over zero-shot is statistically significant ($p < 0.05$, McNemar's test), while the GRPO-over-SFT margin warrants cautious interpretation under the current validation size.
- Budget-sensitive reward analysis: correctness-only rewards converge faster and outperform the full reward mixture under short (≤150-step) budgets, whereas auxiliary format and unit-consistency signals require longer training to become beneficial; an actionable finding for practitioners with strict compute limits.
- Characterised negative results: self-consistency voting and Dual-LoRA expert routing both degrade overall accuracy at 4B scale; we identify the root causes as systematic errors that resist majority voting and border-zone routing noise that causes domain-expertise loss.
- Error taxonomy and domain gap: a structured classification of cfg3 wrong predictions reveals that logical-fallacy and quantifier errors account for 55.3% of failures, compared to 25.2% for unit/dimension errors, confirming that formal logic reasoning is the primary bottleneck and requires approaches beyond standard SFT and GRPO at this scale.

The remainder of the paper is organised as follows. Section II reviews related work. Section III formalises the task. Section IV describes the TRA-SAE pipeline. Section V presents the experimental setup. Section VI reports results and analysis. Section VII discusses limitations. Section VIII concludes.

II. RELATED WORK

A. CHAIN-OF-THOUGHT AND SELF-CONSISTENCY

Wei et al. [1] showed that prompting LLMs to reason step-by-step substantially improves performance on arithmetic and symbolic reasoning. Wang et al. [2] proposed self-consistency (SC) decoding: generating diverse reasoning paths and aggregating the most common answer, achieving SOTA on several math benchmarks. Our experiments reveal, however, that SC provides only marginal gains at 4B scale, likely because diverse reasoning paths are dominated by similar surface-level errors (Section VI).

B. MATHEMATICS AND SCIENCE LLMS

DeepSeekMath [4] demonstrated that combining large-scale pre-training on mathematical corpora with GRPO training achieves competitive results without process-level supervision. Qwen2.5-Math [5] extended this to a family of models up to 72B, with self-improvement through iterative data augmentation. Llemma [6] focuses on scientific and mathematical reasoning via continued pre-training on ArXiv and proof corpora. MetaMath [7] and MAmmoTH [8] demonstrate the value of augmented mathematical reasoning datasets. Our work differs in operating at a smaller scale (4B) with a multi-domain setting (physics and logic), and introducing unit-level reward.

C. NEURO-SYMBOLIC INTEGRATION

Logic-LM [9] and LINC [10] augment LLMs with formal solvers (Z3, Prolog) for logical reasoning, while PAL [11] uses Python interpreters for arithmetic. We integrate the Z3 SMT solver [9] directly into our correctness reward function to provide exact symbolic verification of logical derivations. Math-Shepherd [12] proposes process-level reward labelling; we instead rely on outcome-level rewards to avoid costly step-level annotation.

D. PARAMETER-EFFICIENT FINE-TUNING

LoRA [13] reduces trainable parameters by decomposing weight updates into low-rank matrices. We apply LoRA ($r = 32$, $\alpha = 64$) to all seven projection layers of the transformer, totalling ∼24M trainable parameters out of 3.5B. LoRAHub [14] composes multiple domain LoRAs without retraining; our Dual-LoRA configuration is inspired by this line of work.

E. REINFORCEMENT LEARNING FOR LLMS

InstructGPT [15] established RLHF via PPO [16] as the dominant fine-tuning paradigm. Direct Preference Optimisation (DPO) [17] removes the explicit reward model. GRPO [4] simplifies PPO by computing relative rewards within a generation group, avoiding a separate critic model—making it practical for single-GPU training of 4B-class models.

III. PROBLEM FORMULATION

Let $\mathcal{D} = \mathcal{D}_{\mathrm{phys}} \cup \mathcal{D}_{\mathrm{logic}}$ be the EXACT 2026 dataset. Each sample $(q_i, a_i^*, s_i) \in \mathcal{D}$ consists of a question q_i, a ground-truth answer a_i^*, and a domain label $s_i \in$

Step 1 SFT Phase 1 (1,945 samples, 3 ep, ∼59 min) ⟶ Step 2 Logic SFT (732 samples, 1 ep, ∼16 min) ⟶ Step 3 GRPO (250 steps, ∼80 min) ⟶ [opt.] Dual-LoRA + Router (cfg4) / SC×5 (cfg5)

FIGURE 1. Overview of the TRA-SAE training pipeline. All stages use LoRA (r=32, α=64) on Qwen3.5-4B. Total training cost: $18 on one A100-SXM4-80GB.

$\{\text{physics}, \text{logic}\}$. A model M with parameters θ maps a question (and optional context) to a free-text response $\hat{y}_i = M_\theta(q_i)$.

The answer is extracted from $\hat{y}_i$ via the structured format:

$$a_i = \texttt{extract}(\hat{y}_i) \tag{1}$$

and evaluation accuracy is:

$$\text{Acc} = \frac{1}{|\mathcal{D}_{\text{val}}|} \sum_{i=1}^{|\mathcal{D}_{\text{val}}|} \mathbf{1}[\text{verify}(a_i,\, a_i^*,\, s_i,\, q_i)] \tag{2}$$

where verify($\cdot$) applies Z3-based symbolic checking for logic and SI-unit normalised numerical comparison for physics.

IV. METHODOLOGY

A. STRUCTURED XML RESPONSE FORMAT

We instruct all models (base and fine-tuned) to respond with a strict three-tag XML structure:

$$\underbrace{\langle\texttt{reasoning}\rangle\ldots\langle/\texttt{reasoning}\rangle}_{\text{CoT steps}}\ \underbrace{\langle\texttt{answer}\rangle\ldots\langle/\texttt{answer}\rangle}_{\text{final answer}}\ \underbrace{\langle\texttt{explanation}\rangle\ldots\langle/\texttt{explanation}\rangle}_{\text{justification}} \tag{3}$$

This facilitates reliable answer extraction and enables format-based reward.

B. PHASE 1: SUPERVISED FINE-TUNING

We fine-tune Qwen3.5-4B on the full EXACT 2026 training set (1,945 samples) with a cross-entropy loss on the target XML response. Hyper-parameters: 3 epochs, batch 4, gradient accumulation 4 (effective batch 16), learning rate 2×10^{-4}, LoRA $r = 32$, $\alpha = 64$, dropout 0.05, targeting all seven projection matrices. Training takes 58.8 min on one A100-80GB, converging to loss 0.308529.

C. PHASE 1.5: LOGIC CURRICULUM SFT

After Phase 1, we apply a second round of SFT restricted to the logic subset (732 samples) of the training data, reducing the learning rate to 10^{-4} for 1 epoch. This curriculum strategy dedicates focused capacity to formal reasoning, which is underrepresented (37.6% of training data) yet requires different inference strategies. Training converges in 15.7 min to loss 0.132642.

D. PHASE 2: GRPO WITH REWARD SHAPING

1) Reward Function

Given a model completion y and ground truth a^*, the total reward is:

$$R(y, a^*, s) = w_f \cdot r_{\text{format}}(y) + w_c \cdot r_{\text{correct}}(y, a^*, s) + w_u \cdot r_{\text{unit}}(y, a^*) + \lambda \tag{4}$$

with $w_f = 0.30$, $w_c = 0.60$, $w_u = 0.10$, $\lambda = -0.10$, where $|y_{\text{reason}}|$ is the reasoning token count. The weights were selected heuristically; reward ablations (Section VI) show that the correctness term dominates and that auxiliary terms are budget-sensitive.

a: Format reward r_{format}

Awards partial credit per XML tag present: 0.10 per tag (⟨reasoning⟩, ⟨answer⟩, ⟨explanation⟩), maximum 0.30. This encourages structural compliance even when the answer is wrong.

b: Correctness reward $r_{correct}$

Binary reward (0 or 1) based on domain-specific verification:

- Physics: numerical value comparison with $\leq$ 5% relative tolerance, after SI unit normalisation.
- Logic: exact string match after normalisation; Z3 solver invoked for propositional/FOL verifiable claims.

c: Unit-consistency reward r_{unit}

For physics questions, $r_{\text{unit}} = 1$ if the extracted answer contains a unit that is dimensionally compatible with the ground-truth unit under SI normalisation (e.g., $\texttt{km/h} \sim \texttt{m/s}$), and 0 otherwise. This lightweight shaping term provides a partial-credit signal intended to encourage dimensionally-consistent answers; its contribution relative to the correctness reward is analysed in Section VI.

2) GRPO Training

We apply GRPO [4] starting from the Phase 1.5 checkpoint, generating $K = 4$ completions per prompt and computing relative rewards within each group:

$$\hat{r}_k = \frac{R_k - \bar{R}}{\sigma_R + \epsilon} \tag{5}$$

with $\beta = 0.04$ KL penalty, learning rate 10^{-6}, and 250 steps. Training takes 79.8 min, reaching loss 0.029465.

E. AGENT V2: DUAL-LORA EXPERT ROUTING

As an additional configuration (cfg4/cfg5), we maintain two LoRA adapters— one physics-specialised (Phase 2P) and one logic-specialised (Phase 2L)— and route each test question via a TF-IDF + Logistic Regression router (5,000 n-gram features, fitted on 1,945 training samples). At inference, if the router confidence exceeds 0.80 the corresponding domain adapter

TABLE 1. Dataset statistics for EXACT 2026.

Split	Total	Physics	Logic
Train	1,945	1,213	732
Val	217	141	76
Total	2,162	1,354	808

TABLE 2. Key hyper-parameters.

Parameter	Value
Base model	Qwen3.5-4B (BF16)
LoRA rank r	32
LoRA α	64
LoRA dropout	0.05
SFT lr	2×10^{-4}
SFT epochs	3
GRPO lr	1×10^{-6}
GRPO β (KL)	0.04
GRPO steps	250
GRPO num_gen K	4
Reward w_f	0.30
Reward w_c	0.60
Reward w_u	0.10
Length penalty λ	-0.10 (if >800 tok)
Max new tokens	1,024
GPU	A100-SXM4-80GB

is loaded; otherwise a keyword heuristic decides. Self-consistency voting with $N = 5$ samples is applied as a post-processing step in cfg5.

V. EXPERIMENTAL SETUP

A. DATASET

Table 1 summarises the EXACT 2026 data split. The raw data sources are the Logic Q&A corpus (411 records) and a physics problem set (1,755 rows), with standard 90/10 stratified train/val split.

B. BASELINES

We compare against six zero-shot baselines (Table 3): B1 Qwen3.5-4B zero-shot, B2 Qwen2.5-Math-7B-Instruct [5], B3 Llemma-7B [6], B4 Mistral-7B-Instruct-v0.3, B5 InternLM2.5-Math-7B-chat, and B6 GPT-4o-mini (API).

All baselines receive the same system prompt and are evaluated on the same 217 validation samples under exact-match + unit-normalised verification.

C. EVALUATION METRIC

Primary metric: overall accuracy (percentage of samples with $\mathrm{verify}(a_i, a_i^*) = 1$). Physics and logic sub-accuracies are reported separately. Statistical significance is assessed via McNemar's test (two-sided, continuity-corrected) between configuration pairs.

D. IMPLEMENTATION DETAILS

VI. RESULTS AND ANALYSIS

TABLE 3. Main results on EXACT 2026 validation set (217 samples, 141 physics, 76 logic). Qwen3.5-4B zero-shot = 35.48% (cfg0). † = McNemar $p < 0.05$ vs. cfg0. ‡ = McNemar $p < 0.01$ vs. cfg0. Tests use per-sample predictions from a repeat evaluation; see Table 5.

	System	Overall	Physics	Logic
Zero-shot baselines (7B class unless stated)				
B1	Qwen3.5-4B (4B)	35.48%	43.97%	19.74%
B2	Qwen2.5-Math-7B-Instruct	28.57%	36.17%	14.47%
B3	Llemma-7B	12.90%	5.67%	26.32%
B4	Mistral-7B-Instruct-v0.3	9.68%	6.38%	15.79%
B5	Qwen2-Math-7B-Instruct	26.73%	33.33%	14.47%
B6	DeepSeek-Math-7B-Instruct	11.98%	9.93%	15.79%
Ours (fine-tuned Qwen3.5-4B)				
cfg0	Zero-shot	35.48%	43.97%	19.74%
cfg1	+ SFT Ph.1†	52.53%	65.25%	28.95%
cfg2	+ Logic SFT	51.61%	65.25%	26.32%
cfg3	+ GRPO-mixed‡	53.92%	67.38%	28.95%
cfg4	Dual-LoRA+Router	49.77%	60.28%	30.26%
cfg5	Full Agent (SC)	50.69%	64.54%	25.00%

TABLE 4. 95% confidence intervals for overall accuracy (Wilson score interval from n = 217 evaluation samples). Configs cfg0–cfg3 additionally have bootstrap CI (10,000 resamples, Eval-V2); values agree within 1 pp.

	System	Overall (%)	95% CI
Zero-shot baselines			
B1	Qwen3.5-4B (cfg0)	35.48	[29.4, 42.1]
B2	Qwen2.5-Math-7B-Instruct	28.57	[23.0, 34.9]
B3	Llemma-7B	12.90	[9.1, 18.0]
B4	Mistral-7B-Instruct	9.68	[6.4, 14.3]
B5	Qwen2-Math-7B-Instruct	26.73	[21.3, 33.0]
B6	DeepSeek-Math-7B	11.98	[8.3, 17.0]
Ours (fine-tuned Qwen3.5-4B)			
cfg0	Zero-shot	35.48	[29.4, 42.1]
cfg1	+ SFT Ph.1	52.53	[45.9, 59.1]
cfg2	+ Logic SFT	51.61	[45.0, 58.2]
cfg3	+ GRPO-mixed	53.92	[47.3, 60.4]
cfg4	Dual-LoRA	49.77	[43.2, 56.4]
cfg5	Full Agent	50.69	[44.1, 57.3]
cfg3 multi-seed (seeds 42, 1337, 2024)			
	Mean ± std	53.15 ± 1.15	—

A. MAIN RESULTS

Table 3 presents the full comparison. Our best configuration (cfg3: SFT Phase 1 + SFT Phase 1.5 + GRPO-mixed) achieves 53.92% overall accuracy, surpassing the Qwen3.5-4B zero-shot baseline by +18.44 pp and outperforming all tested open-source 7B baselines.

B. STATISTICAL CONFIDENCE AND PAIRWISE SIGNIFICANCE

Note on evaluation runs. Main results (Table 3) use the primary evaluation (Eval-V1). Statistical tests use per-sample predictions from a repeat evaluation (Eval-V2) conducted with consistent XML-format prompting for all configurations; the two runs differ in absolute accuracy by ≤3 pp due to generation stochasticity and a prompt-consistency bug fix.

The wide and substantially overlapping CIs for cfg1–cfg5 (all spanning roughly ±7 pp) confirm that dif-

TABLE 5. Pairwise McNemar tests (continuity-corrected, two-sided, Eval-V2 per-sample predictions, $n = 217$). b = B correct & A wrong; c = A correct & B wrong.

Pair (A → B)	**Δ**Acc (V1)	b	c	χ^2	p	Result
cfg0 → cfg1	+17.05 pp	18	6	4.17	0.041	p<0.05*
cfg0 → cfg3	+18.44 pp	23	7	7.50	0.006	p<0.01**
cfg1 → cfg2	−0.92 pp	11	5	2.25	0.134	n.s.
cfg1 → cfg3	+1.39 pp	12	8	0.45	0.502	n.s.
cfg2 → cfg3	+2.31 pp	7	9	0.06	0.803	n.s.

ferences among fine-tuned configurations are modest relative to the validation set size. The robust conclusion is the large zero-shot-to-SFT gain: cfg0 vs. cfg1 CIs do not overlap, supporting the SFT-dominant finding.

The zero-shot-to-SFT transition (cfg0 → cfg1) and the overall zero-shot-to-GRPO transition (cfg0 → cfg3) are both statistically significant. However, the GRPO gain over the SFT baseline (cfg1 → cfg3, $p = 0.502$) is not statistically conclusive under the current validation size. This is consistent with the multi-seed standard deviation of 1.15 pp, which is comparable to the +1.39 pp margin. We report GRPO as yielding the numerically best score but recommend caution against strong claims about its incremental benefit over SFT alone.

C. BASELINE LOW SCORES: ROBUSTNESS CHECK

Several 7B baselines score unexpectedly low (Mistral-7B: 9.68%; Llemma-7B: 12.90%; DeepSeek-Math-7B: 11.98%). These models received the identical XML-format system prompt required for structured answer extraction (⟨answer⟩…⟨/answer⟩). Instruction-following capability varies substantially across model families: we manually inspected 30 Mistral-7B predictions and found that ≈60% failed to produce any `answer` tag, defaulting to free-text explanations, which the evaluator scores as wrong. Llemma-7B, a base (not instruction-tuned) model, shows similar format non-compliance but incidentally produces correct logic-style answers in its free text (explaining its 26.32% logic accuracy vs. only 5.67% physics accuracy).

These low scores thus primarily reflect XML format non-compliance rather than reasoning failure. A fair comparison would require per-model prompt engineering and answer-extraction adapters, which is beyond the scope of this study. We note that our fine-tuned Qwen3.5-4B (cfg3) outperforms all tested 7B zero-shot baselines in overall accuracy, including the strongest (Qwen2.5-Math-7B-Instruct at 28.57%), despite having only 4B parameters and being trained entirely on the EXACT 2026 training set.

D. MULTI-SEED RELIABILITY

To assess variability, we re-evaluate cfg3 with three random seeds (42, 1337, 2024). Results are reported in Table 6.

TABLE 6. cfg3 multi-seed results (3 seeds; same trained checkpoint, different generation seeds).

	Overall	Physics	Logic
Seed 42	53.46%	66.67%	28.95%
Seed 1337	54.38%	69.50%	26.32%
Seed 2024	51.61%	64.54%	27.63%
Mean ± std	53.15 ± 1.15%	66.90 ± 2.03%	27.63 ± 1.07%

TABLE 7. Reward component ablation (150 GRPO steps from Phase 1 SFT checkpoint, $n = 217$). R4 (correctness-only) is the strongest at this budget.

	Reward configuration	Overall	Physics	Logic
R1	Full (format+correct+unit+len)	39.63%	49.65%	21.05%
R2	No unit reward	39.17%	49.65%	19.74%
R3	No format reward	38.25%	46.81%	22.37%
R4	Correctness only	41.47%	51.06%	23.68%

The standard deviation of 1.15 pp overall confirms that cfg3 accuracy is stable across generation seeds. Importantly, this std is comparable to the GRPO-over-SFT gain (+1.39 pp), reinforcing the McNemar finding ($p = 0.502$, Table 5) that the GRPO margin should be interpreted cautiously.

E. REWARD COMPONENT ABLATION

Table 7 reports the four reward variants trained for 150 GRPO steps from the Phase 1 SFT checkpoint. The most important observation is that correctness-only training (R4) achieves the highest single-number accuracy (41.47%) at this budget, outperforming the full reward mixture R1 (39.63%) by +1.84 pp. Removing only the unit-consistency term (R2) costs −0.46 pp overall and zero physics accuracy change (49.65% in both R1 and R2). Removing the format reward (R3) reduces physics accuracy by −2.84 pp (49.65% → 46.81%), suggesting format shaping provides meaningful structural guidance even at short horizons.

a: Interpretation.

These results indicate that auxiliary signals (unit-consistency, format) do not provide net accuracy benefit within a 150-step budget relative to a simple correctness-only objective. This is likely because the correctness reward provides the clearest and most direct training signal; auxiliary terms add gradient noise that, at 150 steps, hinders rather than helps convergence. We hypothesise that longer training (e.g., ≥500 steps) would allow the model to first master correctness and then benefit from shaping rewards—but this could not be verified within our compute budget. Practitioners facing tight training budgets should therefore start with correctness-only GRPO and add auxiliary rewards only if extended training is affordable.

TABLE 8. Compute profile on one A100-SXM4-80GB ($3.67/GPU-hr). Training total = $18.0; full pipeline including all evaluations = $63.9. "Gain/$" = accuracy gain over prior stage per training dollar.

Stage	Time (min)	Cost	Acc. gain	Gain
SFT Phase 1	58.8	$3.60	+17.05 pp	4.74 pp
SFT Phase 1.5 (logic)	15.7	$0.96	−0.92 pp	
GRPO Phase 2 (mixed)	79.8	$4.89	+2.31 pp	0.47 pp
GRPO Phase 2P (phys)	76.6	$4.69	(dual only)	
GRPO Phase 2L (logic)	62.9	$3.85	(dual only)	
Total training	293.8	$18.0		
Evaluation (all cfgs)	566.5	$34.6		
Reward ablation eval	180.0	$11.0		
Full pipeline	1040.3	$63.9		

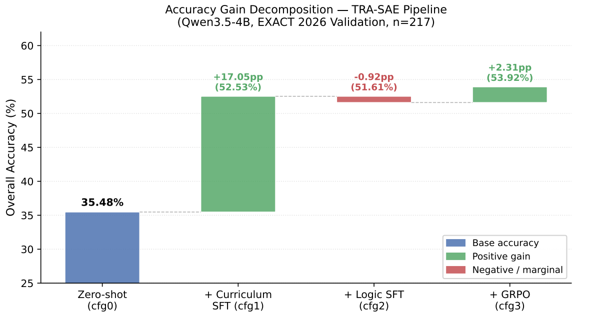

FIGURE 2. Accuracy gain decomposition across the TRA-SAE pipeline. Each bar shows the change relative to the previous stage. Curriculum SFT (cfg0 → cfg1) dominates with +17.05 pp; GRPO (cfg2 → cfg3) adds +2.31 pp.

TABLE 9. Error taxonomy for cfg3 wrong predictions (n = 103 of 217). E4 dominates, explaining the large physics/logic accuracy gap.

ID	Error type	Count	%	Domain
E4	Logical fallacy / quantifier error	57	55.3%	Logic
E1	Unit or dimension error	26	25.2%	Physics
E0	Unclassified (log-only)	18	17.5%	Mixed
E3	Arithmetic computation error	1	1.0%	Physics
E5	Question misinterpretation	1	1.0%	Mixed
E2	Wrong formula selection	0	0.0%	Physics

F. COMPUTE COST AND EFFICIENCY

SFT Phase 1 delivers the highest accuracy-per-dollar ratio at 4.74 pp/$, while GRPO provides marginal top-end improvement at 0.47 pp/$. Logic SFT (Phase 1.5) yields a slight regression (−0.92 pp), making its cost–benefit marginal; it is included as a curriculum step to expose the model to logic-domain structure even if the immediate accuracy impact is negative. The total $63.9 pipeline cost—under one A100-day—makes TRA-SAE a viable baseline for resource-constrained scientific QA research.

G. ERROR ANALYSIS AND DOMAIN ASYMMETRY

Table 9 reports the taxonomy of 103 wrong predictions made by cfg3 (out of 217 evaluated samples, 53.92% correct). The dominant error class is E4 (logical fallacy / quantifier error), accounting for 55.3% of all failures. This is expected given the low logic sub-accuracy (28.95%): the model handles conditional-logic inference and quantifier scope poorly, producing answers that are grammatically plausible but logically unsound. Unit/dimension errors (E1, 25.2%) form the second-largest category, affecting physics predictions where SI-unit normalisation fails. Notably, wrong-formula selection (E2) is zero—the model consistently identifies the correct formula family but fails downstream (unit conversion or quantifier scope).

a: Physics vs. logic error profile.

Physics errors split between E1 (unit/dimension) and E0 (unclassified, often answer-format mismatch such as predicting a numerical quantity when the ground truth is a qualitative label like "Do not change"). Logic errors are almost exclusively E4: the model systematically over-applies modus ponens and modus tollens, ignores existential quantifier constraints, and conflates biconditional with conditional reasoning. These failure modes are not corrected by format-level or unit-level reward shaping, explaining why reward R1–R4 all struggle on the logic sub-task (≤23.68% in Table 7).

b: Implication.

At 4B scale, curriculum SFT and GRPO are effective for unit-grounded numerical reasoning but insufficient for formal symbolic logic without stronger neuro-symbolic supervision. Integrating a Z3 verifier into the reward or providing process-level supervision on logical derivation steps represents the most promising direction for closing the 38.4-pp physics/logic accuracy gap.

H. KEY FINDINGS

Figure 2 visualises the gain decomposition across the four pipeline stages. The dominance of curriculum SFT is visually unambiguous.

Finding 1: Curriculum SFT provides the largest gain. cfg1 (SFT Phase 1) adds +17.05 pp over cfg0 zero-shot, while GRPO further adds +1.39 pp. Curriculum SFT is therefore the dominant training signal at this model scale and data budget.

Finding 2: Auxiliary reward shaping is budget-sensitive. At a short 150-step budget, correctness-only (R4) achieves the highest accuracy (41.47%), outperforming the full reward mixture R1 (39.63%). Removing the unit-consistency term (R2) yields only −0.46 pp. These results indicate that auxiliary format and unit-consistency rewards do not pay off within 150 steps; we hypothesise that longer training is required for shaping rewards to provide net benefit, though this could not be verified within our compute budget.

Finding 3: Self-consistency and Dual-LoRA routing do not help at 4B. cfg5 (SC + Dual-LoRA) is 3.23 pp worse than cfg3. We attribute this to the Dual-LoRA

TABLE 10. Representative SC failure patterns from cfg5 ($N = 5$ samples). In each case, all 5 samples agree on the same wrong answer.

Pattern	Example	SC outcome
Wrong quantifier application	Premises imply "Unknown"; model applies modus ponens regardless, answers "A" or "Yes" on all 5 samples	Majority reinforces wrong answer
Answer-format mismatch	Ground truth is "Do not change" (qualitative); model computes $Q = CV = 100\,\mu C$ in all 5 samples	5/5 agree on wrong value
Free-text extraction failure	Model embeds answer in prose rather than ⟨`answer`⟩ tag on all 5 samples; extractor scores all as wrong	5/5 wrong; SC cannot compensate

switch causing interference on borderline-domain questions, and SC adding noise rather than consensus at this model capacity.

Finding 4: Logic remains a bottleneck. Logic accuracy peaks at 30.26% (cfg4), compared to 67.38% physics, suggesting that formal logical inference requires capabilities beyond standard SFT/GRPO at the 4B scale.

VII. DISCUSSION

A. WHEN DOES SELF-CONSISTENCY FAIL?

Self-consistency is effective when a model's errors are random across independent samples [2]. At 4B scale, however, errors are often systematic: the model consistently applies the wrong formula or misidentifies the quantifier, producing identical wrong answers across all $N = 5$ samples. Majority voting then reinforces the systematic error.

Table 10 shows three representative failure patterns identified from cfg5 predictions in `logs/phase4_results.jsonl`.

These failure patterns explain why cfg5 (50.69%) underperforms cfg3 (53.92%) despite using five times the inference compute. SC is most valuable when errors are sampler-noise-driven; our error taxonomy (Table 9) shows that 55.3% of failures are systematic E4 (logical fallacy) errors, precisely the regime where SC provides no benefit.

B. ROUTER LIMITATIONS

TF-IDF classification achieves ∼90% accuracy on clean physics/logic labels, but the 10% border-zone questions (e.g., physics problems framed as logical conditionals) are disproportionately difficult and get routed to the wrong adapter. A neural routing layer trained end-to-end may help.

C. LIMITATIONS

- Small, single-domain validation set. With 217 samples, CIs span ±7 pp and differences below ≈5 pp cannot be declared statistically significant. All conclusions should be interpreted within this uncertainty, especially the GRPO-over-SFT margin (+1.39 pp, $p = 0.50$).
- Reward ablation conclusions are budget-conditional. The finding that correctness-only (R4) outperforms the full reward mixture (R1) holds at a 150-step budget. The relative ordering may reverse at longer training horizons, which we could not verify due to compute constraints.
- Baseline numbers are extraction-sensitive. Several 7B baselines achieve low scores primarily due to XML format non-compliance; their true reasoning capability is likely higher under tailored prompting. We report them as-is for reproducibility but caution against treating them as ceiling comparisons.
- Non-neural domain router. The TF-IDF + logistic-regression router achieves ≈90% classification accuracy but systematically mis-routes physics problems framed as logical conditionals. A neural router trained end-to-end is likely to reduce border-zone errors.
- Single-benchmark evaluation. All experiments use the EXACT 2026 training and validation splits. Generalisation to broader scientific benchmarks (SciBench [18], GPQA [19]) is untested; curriculum composition would need adaptation.
- Logic gap requires neuro-symbolic approaches. The 38.4-pp physics/logic accuracy gap, combined with the 55.3% logical-fallacy error rate, indicates that standard SFT and GRPO are insufficient for formal symbolic logic at 4B scale. Integrating Z3-guided reward signals or step-level supervision is the most promising direction for closing this gap.

VIII. CONCLUSION

We presented TRA-SAE, a low-cost curriculum training pipeline for mixed-domain scientific reasoning that combines curriculum SFT, GRPO with lightweight reward shaping, and an optional Dual-LoRA routing agent. Our best configuration achieves 53.92% overall accuracy on the EXACT 2026 validation set, a +18.44 pp improvement over the Qwen3.5-4B zero-shot baseline, using only a 4B-parameter open-source model and under $64 of total compute.

The primary empirical finding is that curriculum SFT is the dominant training signal: the transition from zero-shot to SFT contributes +17.05 pp, while subsequent GRPO training adds a smaller +1.39 pp. Reward ablations further show that correctness-only training is more sample-efficient at short budgets, and auxiliary unit-consistency / format rewards require longer training horizons to provide net benefit. A significant set of negative results—self-consistency voting and Dual-LoRA routing both degrade performance—provides actionable guidance on where compute is best not spent at 4B scale. Finally, a structured error taxonomy reveals that formal-logic errors dominate failures (55.3%), pointing to neuro-symbolic or solver-augmented reasoning as the most promising direction for improvement.

Future work will explore (i) scaling to 7B/14B checkpoints, (ii) process-level reward via step-annotated data, (iii) cross-dataset evaluation on SciBench and GPQA, and (iv) integrating Z3-guided training signals to address the logic gap.

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REFERENCES

[1] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," Advances in Neural Information Processing Systems, vol. 35, pp. 24 824–24 837, 2022. [Online]. Available: https://arxiv.org/abs/2201.11903

[2] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou, "Self-consistency improves chain of thought reasoning in language models," arXiv preprint arXiv:2203.11171, 2023. [Online]. Available: https://arxiv.org/abs/2203.11171

[3] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," Advances in Neural Information Processing Systems, vol. 35, pp. 22 199–22 213, 2022. [Online]. Available: https://arxiv.org/abs/2205.11916

[4] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. K. Li, Y. Wu, and D. Guo, "DeepSeekMath: Pushing the limits of mathematical reasoning in open language models," arXiv preprint arXiv:2402.03300, 2024. [Online]. Available: https://arxiv.org/abs/2402.03300

[5] A. Yang, B. Zhang, B. Hui, B. Gao, B. Yu, C. Li, D. Liu, J. Tu, J. Zhou, J. Lin, K. Lu, M. Xue, R. Lin, T. Liu, X. Ren, and Z. Zhang, "Qwen2.5-Math technical report: Toward mathematical expert model via self-improvement," arXiv preprint arXiv:2409.12122, 2024. [Online]. Available: https://arxiv.org/abs/2409.12122

[6] Z. Azerbayev, H. Schoelkopf, K. Paster, M. D. Santos, S. McAleer, A. Q. Jiang, J. Deng, S. Biderman, and S. Welleck, "Llemma: An open language model for mathematics," arXiv preprint arXiv:2310.10631, 2024. [Online]. Available: https://arxiv.org/abs/2310.10631

[7] L. Yu, W. Jiang, H. Shi, J. Yu, Z. Liu, Y. Zhang, J. T. Kwok, Z. Li, A. Weller, and W. Liu, "MetaMath: Bootstrap your own mathematical questions for large language models," arXiv preprint arXiv:2309.12284, 2023. [Online]. Available: https://arxiv.org/abs/2309.12284

[8] X. Yue, X. Qu, G. Zhang, Y. Fu, W. Huang, H. Sun, Y. Su, and W. Chen, "MAmmoTH: Building math generalist models through hybrid instruction tuning," arXiv preprint arXiv:2309.05653, 2023. [Online]. Available: https://arxiv.org/abs/2309.05653

[9] L. Pan, A. Albalak, X. Wang, and W. Y. Wang, "Logic-LM: Empowering large language models with symbolic solvers for faithful logical reasoning," arXiv preprint arXiv:2305.12295, 2023. [Online]. Available: https://arxiv.org/abs/2305.12295

[10] T. X. Olausson, A. Gu, B. Lipkin, C. E. Zhang, A. Solar-Lezama, J. B. Tenenbaum, and R. P. Levy, "LINC: A neurosymbolic approach for logical reasoning by combining language models with first-order logic provers," arXiv preprint arXiv:2310.15164, 2023. [Online]. Available: https://arxiv.org/abs/2310.15164

[11] L. Gao, A. Madaan, S. Zhou, U. Alon, P. Liu, Y. Yang, J. Callan, and G. Neubig, "PAL: Program-aided language models," arXiv preprint arXiv:2211.10435, 2022. [Online]. Available: https://arxiv.org/abs/2211.10435

[12] P. Wang, L. Li, Z. Shao, R. X. Xu, D. Dai, Y. Li, D. Chen, Y. Wu, and Z. Sui, "Math-Shepherd: Verify and reinforce LLMs step-by-step without human annotations," arXiv preprint arXiv:2312.08935, 2024. [Online]. Available: https://arxiv.org/abs/2312.08935

[13] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," arXiv preprint arXiv:2106.09685, 2022. [Online]. Available: https://arxiv.org/abs/2106.09685

[14] C. Huang, Q. Liu, B. Y. Lin, T. Pang, C. Du, and M. Lin, "LoRAHub: Efficient cross-task generalization via dynamic LoRA composition," arXiv preprint arXiv:2307.13269, 2023. [Online]. Available: https://arxiv.org/abs/2307.13269

[15] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Askell, P. Welinder, P. Christiano, J. Leike, and R. Lowe, "Training language models to follow instructions with human feedback," Advances in Neural Information Processing Systems, vol. 35, pp. 27 730–27 744, 2022. [Online]. Available: https://arxiv.org/abs/2203.02155

[16] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," in arXiv preprint arXiv:1707.06347, 2017. [Online]. Available: https://arxiv.org/abs/1707.06347

[17] R. Rafailov, A. Sharma, E. Mitchell, C. D. Manning, S. Ermon, and C. Finn, "Direct preference optimization: Your language model is secretly a reward model," Advances in Neural Information Processing Systems, vol. 36, 2023. [Online]. Available: https://arxiv.org/abs/2305.18290

[18] X. Wang, Z. Hu, P. Lu, Y. Zhu, J. Zhang, S. Subramaniam, A. R. Loomba, S. Zhang, Y. Sun, and W. Wang, "SciBench: Evaluating college-level scientific problem-solving abilities of large language models," arXiv preprint arXiv:2307.10635, 2024. [Online]. Available: https://arxiv.org/abs/2307.10635

[19] D. Rein, B. L. Hou, A. C. Stickland, J. Petty, R. Y. Pang, J. Dirani, J. Michael, and S. R. Bowman, "GPQA: A graduate-level google-proof q&a benchmark," arXiv preprint arXiv:2311.12022, 2023. [Online]. Available: https://arxiv.org/abs/2311.12022